```
import pandas as pd
df = pd.read_csv("Base_de_Datos_Eng_5_2_2026_V1.csv", sep=';',
encoding='iso-8859-1')
df
```

```
     Transformer  Measurement Number Power (MVA)    Tp    Ts
Temperature Rise  \
0              T1                 1.0          50   NaN   NaN

1              T1                 2.0          50   NaN   NaN
NaN
2              T2                 1.0           7  69.0  34.5
55
3              T2                 2.0           7  69.0  34.5
55
4              T2                 3.0           7  69.0  34.5
55
...           ...                 ...         ...   ...   ...
...
1861         NaN                 NaN         NaN   NaN   NaN
NaN
1862         NaN                 NaN         NaN   NaN   NaN
NaN
1863         NaN                 NaN         NaN   NaN   NaN
NaN
1864         NaN                 NaN         NaN   NaN   NaN
NaN
1865         NaN                 NaN         NaN   NaN   NaN
NaN

        Year  Impedance (kV)  Hydrogen  Methane  ...  \
0        NaN             NaN      24.0        1  ...
1        NaN             NaN       0.0        0  ...
2     1998.0            9.67       5.0       10  ...
3     1998.0            9.67       0.0       27  ...
4     1998.0            9.67      81.0       13  ...
...      ...             ...       ...      ...  ...
1861     NaN             NaN       6.0        1  ...
1862     NaN             NaN       2.0        1  ...
1863     NaN             NaN       4.0        2  ...
1864     NaN             NaN       0.0        3  ...
1865     NaN             NaN       5.0        2  ...

      Hydroxymethyl Furan (ppb)  Furfuryl Alcohol (ppb)  Furfural
(ppb)  \
0                        ND,<10                  ND,<10
ND,<10
1                        ND,<10                  ND,<10
ND,<10
```

```
2                          ND,<10                   ND,<10
21
3                          ND,<10                   ND,<10
ND,<10
4                          ND,<10                   ND,<10
ND,<10
...                           ...                      ...              ..
.
1861                          NaN                      NaN
NaN
1862                          NaN                      NaN
NaN
1863                          NaN                      NaN
NaN
1864                          NaN                      NaN
NaN
1865                          NaN                      NaN
NaN

      Acetyl Furan (ppb)  Methyl Furfural (ppb)  Diagnosis by Basic
Gas Ratio  \
0                  ND,<10                   ND,<10
Undefined
1                  ND,<10                   ND,<10
Undefined
2                  ND,<10                   ND,<10   Thermal fault, >700
°C (T3)
3                  ND,<10                   ND,<10
Undefined
4                  ND,<10                   ND,<10
Undefined
...                   ...                      ...
...
1861                  NaN                      NaN
Undefined
1862                  NaN                      NaN
Undefined
1863                  NaN                      NaN
Undefined
1864                  NaN                      NaN
Undefined
1865                  NaN                      NaN
Unidentifiable Fault

      Diagnosis by Dornenburg         Diagnosis by Rogers  \
0        Unidentifiable Fault                    Undefined
1        Unidentifiable Fault                    Undefined
2        Unidentifiable Fault  Thermal fault, >700°C (T3)
3        Unidentifiable Fault                    Undefined
```

```
4        Unidentifiable Fault                    Undefined
...                       ...                          ...
1861     Unidentifiable Fault                    Undefined
1862     Unidentifiable Fault                    Undefined
1863     Unidentifiable Fault                    Undefined
1864     Unidentifiable Fault                    Undefined
1865     Unidentifiable Fault         Unidentifiable Fault

                          Diagnosis by Duval  Unnamed: 47
0     Thermal fault,  >300 °C - <700 °C (T2)          NaN
1                                 Undefined          NaN
2                       Unidentifiable Fault          NaN
3     Thermal fault,  >300 °C - <700 °C (T2)          NaN
4                       Unidentifiable Fault          NaN
...                                      ...          ...
1861                    Unidentifiable Fault          NaN
1862              Thermal fault, >700°C (T3)          NaN
1863              Thermal fault, >700°C (T3)          NaN
1864              Thermal fault, >700°C (T3)          NaN
1865              Thermal fault, >700°C (T3)          NaN

[1866 rows x 48 columns]
```

Notes: Gasses have perfect completeness. Other paramaters have low completion rates. There may be limited use in imputating gases Many of these features are strings and need to be converted into a numerical format. To be considered next week

```
completeness_percent = (1866 - df.isnull().sum())/1866
print("Data 'completeness' percentage:")
print(completeness_percent)
print(df.dtypes)
```

```
Data 'completeness' percentage:
Transformer                                                 0.500000
Measurement Number                                          0.500000
Power (MVA)                                                 0.500000
Tp                                                          0.498928
Ts                                                          0.498928
Temperature Rise                                            0.392283
Year                                                        0.432476
Impedance (kV)                                              0.421222
Hydrogen                                                    1.000000
Methane                                                     1.000000
Ethane                                                      1.000000
Ethylene                                                    1.000000
Acetylene                                                   1.000000
Carbon Monoxide                                             1.000000
Carbon Dioxide                                              1.000000
Nitrogen                                                    1.000000
```

```
Oxygen                                                    1.000000
Total Dissolved Combustible Gases                         0.995713
TDCG (ppm)                                                0.995713
Sample Temperature                                        0.492497
% Moisture by Dry Weight of Cellulose Insulation          0.174169
% Relative Water Saturation in Oil                        0.215434
% Relative Humidity Saturation                            0.258842
Moisture Content (ppm)                                    0.499464
Neutralization Number (mg KOH/g)                          0.499464
Inhibitor Content ()                                      0.335477
Corrosive Sulfur Test                                     0.029475
Color                                                     0.499464
Visual Inspection                                         0.498928
Viscosity (cSt) at 25 °C                                  0.385316
Viscosity (SSU) at 25 °C                                  0.378885
Specific Gravity (60/60 °F)                               0.431940
Interfacial Tension (dynes/cm)                            0.499464
Power Factor at 25 °C                                     0.499464
Power Factor at 100 °C                                    0.461951
Dielectric Breakdown (kV) at 25 °C                        0.390675
Dielectric Breakdown (kV) at 1.016 mm (0.04 in) at 25 °C  0.494641
Dielectric Breakdown (kV) at 2.032 mm (0.08 in) at 25 °C  0.351554
Hydroxymethyl Furan (ppb)                                 0.481779
Furfuryl Alcohol (ppb)                                    0.480171
Furfural (ppb)                                            0.476420
Acetyl Furan (ppb)                                        0.478564
Methyl Furfural (ppb)                                     0.477492
Diagnosis by Basic Gas Ratio                              1.000000
Diagnosis by Dornenburg                                   1.000000
Diagnosis by Rogers                                       1.000000
Diagnosis by Duval                                        1.000000
Unnamed: 47                                               0.000536
dtype: float64
Transformer                                                  str
Measurement Number                                       float64
Power (MVA)                                                  str
Tp                                                       float64
Ts                                                       float64
Temperature Rise                                             str
Year                                                     float64
Impedance (kV)                                           float64
Hydrogen                                                 float64
Methane                                                    int64
Ethane                                                   float64
Ethylene                                                   int64
Acetylene                                                  int64
Carbon Monoxide                                            int64
Carbon Dioxide                                             int64
Nitrogen                                                   int64
```

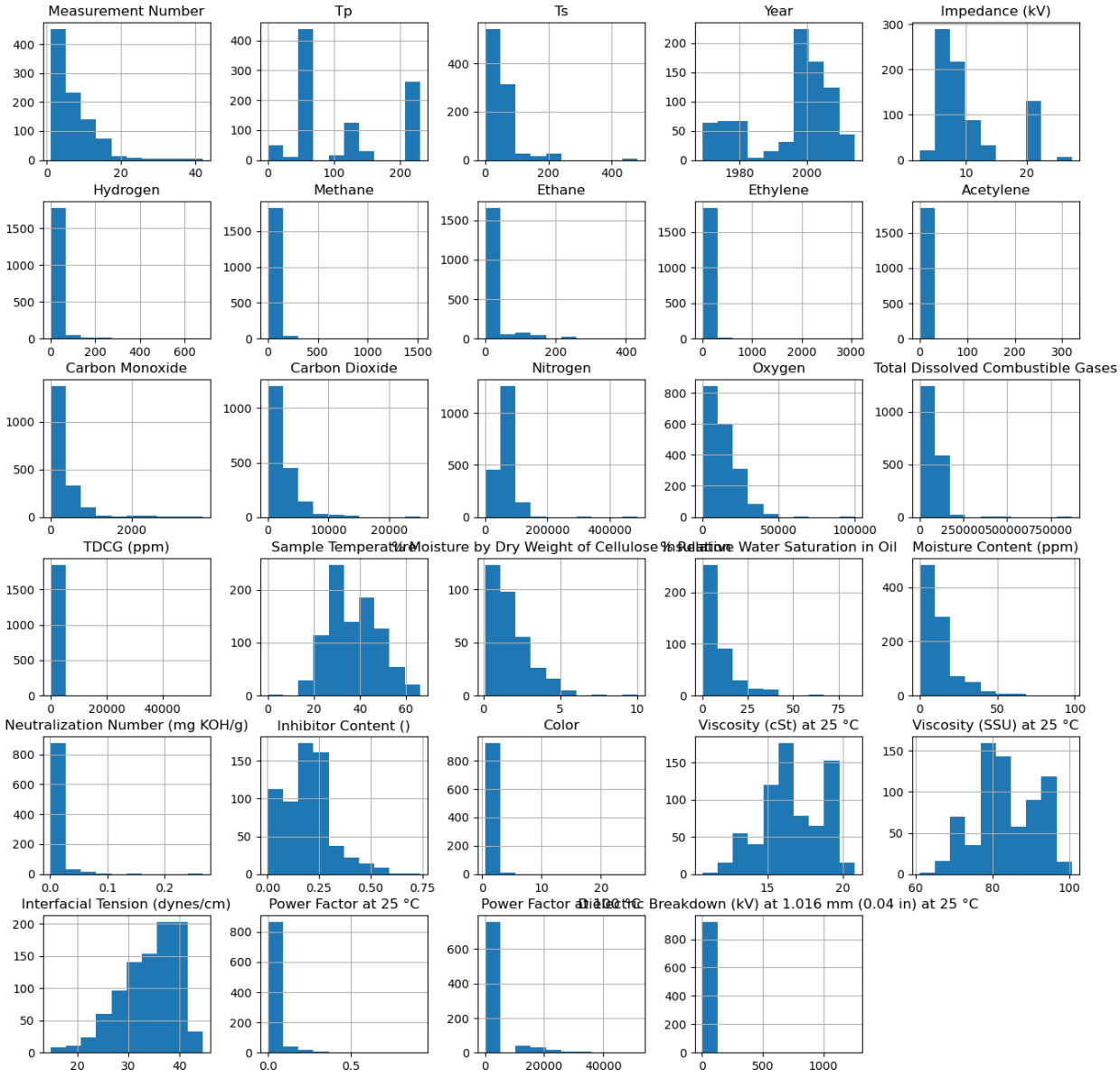
```
Oxygen                                                          int64
Total Dissolved Combustible Gases                             float64
TDCG (ppm)                                                    float64
Sample Temperature                                            float64
% Moisture by Dry Weight of Cellulose Insulation              float64
% Relative Water Saturation in Oil                            float64
% Relative Humidity Saturation                                    str
Moisture Content (ppm)                                        float64
Neutralization Number (mg KOH/g)                              float64
Inhibitor Content ()                                          float64
Corrosive Sulfur Test                                             str
Color                                                         float64
Visual Inspection                                                 str
Viscosity (cSt) at 25 °C                                      float64
Viscosity (SSU) at 25 °C                                      float64
Specific Gravity (60/60 °F)                                       str
Interfacial Tension (dynes/cm)                                float64
Power Factor at 25 °C                                         float64
Power Factor at 100 °C                                        float64
Dielectric Breakdown (kV) at 25 °C                                str
Dielectric Breakdown (kV) at 1.016 mm (0.04 in) at 25 °C      float64
Dielectric Breakdown (kV) at 2.032 mm (0.08 in) at 25 °C          str
Hydroxymethyl Furan (ppb)                                         str
Furfuryl Alcohol (ppb)                                            str
Furfural (ppb)                                                    str
Acetyl Furan (ppb)                                                str
Methyl Furfural (ppb)                                             str
Diagnosis by Basic Gas Ratio                                      str
Diagnosis by Dornenburg                                           str
Diagnosis by Rogers                                               str
Diagnosis by Duval                                                str
Unnamed: 47                                                       str
dtype: object
```

```
fig = plt.figure(figsize=(16,16))
ax = fig.gca()
df.hist(ax=ax)
```

```
C:\Users\sonvi\AppData\Local\Temp\ipykernel_48676\1608841188.py:3:
UserWarning: To output multiple subplots, the figure containing the
passed axes is being cleared.
  df.hist(ax=ax)
```

```
array([[<Axes: title={'center': 'Measurement Number'}>,
        <Axes: title={'center': 'Tp'}>, <Axes: title={'center':
'Ts'}>,
        <Axes: title={'center': 'Year'}>,
        <Axes: title={'center': 'Impedance (kV)'}>],
       [<Axes: title={'center': 'Hydrogen'}>,
        <Axes: title={'center': 'Methane'}>,
```

```
        <Axes: title={'center': 'Ethane'}>,
        <Axes: title={'center': 'Ethylene'}>,
        <Axes: title={'center': 'Acetylene'}>],
       [<Axes: title={'center': 'Carbon Monoxide'}>,
        <Axes: title={'center': 'Carbon Dioxide'}>,
        <Axes: title={'center': 'Nitrogen'}>,
        <Axes: title={'center': 'Oxygen'}>,
        <Axes: title={'center': 'Total Dissolved Combustible
Gases'}>],
       [<Axes: title={'center': 'TDCG (ppm)'}>,
        <Axes: title={'center': 'Sample Temperature'}>,
        <Axes: title={'center': '% Moisture by Dry Weight of Cellulose
Insulation'}>,
        <Axes: title={'center': '% Relative Water Saturation in
Oil'}>,
        <Axes: title={'center': 'Moisture Content (ppm)'}>],
       [<Axes: title={'center': 'Neutralization Number (mg KOH/g)'}>,
        <Axes: title={'center': 'Inhibitor Content ()'}>,
        <Axes: title={'center': 'Color'}>,
        <Axes: title={'center': 'Viscosity (cSt) at 25 °C'}>,
        <Axes: title={'center': 'Viscosity (SSU) at 25 °C'}>],
       [<Axes: title={'center': 'Interfacial Tension (dynes/cm)'}>,
        <Axes: title={'center': 'Power Factor at 25 °C'}>,
        <Axes: title={'center': 'Power Factor at 100 °C'}>,
        <Axes: title={'center': 'Dielectric Breakdown (kV) at 1.016 mm
(0.04 in) at 25 °C'}>,
        <Axes: >]], dtype=object)
```




```
corr = df.select_dtypes(include='number').corr()
corr
```

	Measurement Number
Measurement Number	1.000000
Tp	-0.146860
Ts	-0.128407
Year	-0.277138
Impedance (kV)	-0.316714

```
Hydrogen                                                    -0.042550
Methane                                                     -0.035415
Ethane                                                      -0.009418
Ethylene                                                     0.016463
Acetylene                                                   -0.007985
Carbon Monoxide                                             -0.025091
Carbon Dioxide                                               0.125651
Nitrogen                                                     0.011044
Oxygen                                                       0.090905
Total Dissolved Combustible Gases                            0.043810
TDCG (ppm)                                                  -0.026598
Sample Temperature                                           0.001532
% Moisture by Dry Weight of Cellulose Insulation            -0.038273
% Relative Water Saturation in Oil                          -0.052896
Moisture Content (ppm)                                       0.088996
Neutralization Number (mg KOH/g)                             0.131683
Inhibitor Content ()                                        -0.028936
Color                                                        0.102474
Viscosity (cSt) at 25 °C                                     0.000214
Viscosity (SSU) at 25 °C                                    -0.000611
Interfacial Tension (dynes/cm)                              -0.150736
Power Factor at 25 °C                                        0.030183
Power Factor at 100 °C                                       0.096063
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...          -0.013980

                                                                 Tp        Ts
\
```

Measurement Number	-0.146860	-0.128407
Tp	1.000000	0.211249
Ts	0.211249	1.000000
Year	-0.030244	-0.161123
Impedance (kV)	0.778691	0.457659
Hydrogen	-0.001746	0.040537
Methane	-0.068295	-0.059837
Ethane	-0.177437	-0.164110
Ethylene	-0.040344	-0.033251
Acetylene	-0.011061	-0.014960
Carbon Monoxide	0.037381	0.002172
Carbon Dioxide	-0.157295	-0.062152
Nitrogen	-0.072799	-0.031831
Oxygen	0.026157	0.048727
Total Dissolved Combustible Gases	-0.079634	-0.048794
TDCG (ppm)	-0.037409	-0.021451
Sample Temperature	-0.122681	-0.159134
% Moisture by Dry Weight of Cellulose Insulation	0.003329	0.099614
% Relative Water Saturation in Oil	0.049492	0.199839
Moisture Content (ppm)	-0.035007	0.218712
Neutralization Number (mg KOH/g)	-0.063318	0.096555
Inhibitor Content ()	-0.042587	-0.070982
Color	0.015721	0.014208
Viscosity (cSt) at 25 °C	0.185938	-0.078983
Viscosity (SSU) at 25 °C	0.184051	-0.079440
Interfacial Tension (dynes/cm)	0.219579	-0.004872

```
Power Factor at 25 °C                                0.012373  0.013883

Power Factor at 100 °C                               0.194837  0.102022

Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...  0.007745 -0.018111


                                                          Year
Impedance (kV)  \
Measurement Number                                  -0.277138       -
0.316714
Tp                                                  -0.030244
0.778691
Ts                                                  -0.161123
0.457659
Year                                                 1.000000
0.069973
Impedance (kV)                                       0.069973
1.000000
Hydrogen                                             0.033790       -
0.069826
Methane                                              0.065476       -
0.006934
Ethane                                              -0.007182       -
0.141146
Ethylene                                            -0.011863       -
0.010428
Acetylene                                            0.032949       -
0.015477
Carbon Monoxide                                     -0.045637
0.142275
Carbon Dioxide                                      -0.222712       -
0.116658
Nitrogen                                            -0.070159       -
0.044984
Oxygen                                              -0.024015       -
0.110598
Total Dissolved Combustible Gases                   -0.101877       -
0.065445
TDCG (ppm)                                           0.046606
0.085760
Sample Temperature                                  -0.116092       -
0.161740
% Moisture by Dry Weight of Cellulose Insulation    -0.297200       -
0.116638
% Relative Water Saturation in Oil                  -0.215229
0.045948
Moisture Content (ppm)                              -0.331950       -
0.077189
Neutralization Number (mg KOH/g)                    -0.375845       -
```

```
0.054531
Inhibitor Content ()                                -0.200802       -
0.210278
Color                                               -0.337180
0.024532
Viscosity (cSt) at 25 °C                             0.168518
0.115061
Viscosity (SSU) at 25 °C                             0.173908
0.112102
Interfacial Tension (dynes/cm)                       0.359911
0.152328
Power Factor at 25 °C                               -0.175599
0.033291
Power Factor at 100 °C                              -0.412303
0.280679
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...   0.042264
0.000164
```

	Hydrogen	Methane
Measurement Number	-0.042550	-0.035415
Tp	-0.001746	-0.068295
Ts	0.040537	-0.059837
Year	0.033790	0.065476
Impedance (kV)	-0.069826	-0.006934
Hydrogen	1.000000	0.546736
Methane	0.546736	1.000000
Ethane	0.169956	0.511888
Ethylene	0.547985	0.924627
Acetylene	0.586309	0.870122
Carbon Monoxide	0.166605	0.263224
Carbon Dioxide	0.079640	0.089160
Nitrogen	0.104301	0.083903
Oxygen	0.002934	-0.184370
Total Dissolved Combustible Gases	0.031040	0.026382
TDCG (ppm)	0.129876	0.178927

Sample Temperature	0.012054	0.022012
% Moisture by Dry Weight of Cellulose Insulation	0.078584	-0.043986
% Relative Water Saturation in Oil	0.050101	-0.044630
Moisture Content (ppm)	0.091707	-0.038066
Neutralization Number (mg KOH/g)	0.020991	-0.015235
Inhibitor Content ()	0.068996	-0.020633
Color	-0.031327	0.029960
Viscosity (cSt) at 25 °C	0.014048	0.029116
Viscosity (SSU) at 25 °C	0.012675	0.026225
Interfacial Tension (dynes/cm)	-0.040029	-0.063843
Power Factor at 25 °C	0.087803	0.047159
Power Factor at 100 °C	-0.016694	0.051519
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...	-0.015877	0.010466

	Ethane	Ethylene
\		
Measurement Number	-0.009418	0.016463
Tp	-0.177437	-0.040344
Ts	-0.164110	-0.033251
Year	-0.007182	-0.011863
Impedance (kV)	-0.141146	-0.010428
Hydrogen	0.169956	0.547985
Methane	0.511888	0.924627
Ethane	1.000000	0.304074
Ethylene	0.304074	1.000000
Acetylene	0.255992	0.914943
Carbon Monoxide	0.034982	0.281634

Carbon Dioxide	0.274614	0.059609
Nitrogen	0.152551	0.046638
Oxygen	-0.299281	-0.086929
Total Dissolved Combustible Gases	0.114648	0.021640
TDCG (ppm)	0.059428	0.184657
Sample Temperature	0.088886	-0.011757
% Moisture by Dry Weight of Cellulose Insulation	-0.003482	-0.023678
% Relative Water Saturation in Oil	-0.020549	-0.033211
Moisture Content (ppm)	-0.027405	-0.026581
Neutralization Number (mg KOH/g)	-0.018092	0.035468
Inhibitor Content ()	-0.162704	0.041194
Color	0.063606	0.069724
Viscosity (cSt) at 25 °C	0.044893	-0.014452
Viscosity (SSU) at 25 °C	0.040317	-0.016951
Interfacial Tension (dynes/cm)	-0.159603	-0.104643
Power Factor at 25 °C	0.054672	0.038914
Power Factor at 100 °C	0.032568	0.114889
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...	0.050282	-0.011526

	Acetylene	...	\
Measurement Number	-0.007985	...	
Tp	-0.011061	...	
Ts	-0.014960	...	
Year	0.032949	...	
Impedance (kV)	-0.015477	...	
Hydrogen	0.586309	...	
Methane	0.870122	...	
Ethane	0.255992	...	
Ethylene	0.914943	...	
Acetylene	1.000000	...	
Carbon Monoxide	0.230513	...	
Carbon Dioxide	0.014282	...	
Nitrogen	0.052093	...	

```
Oxygen                                                -0.044736  ...
Total Dissolved Combustible Gases                      0.026339  ...
TDCG (ppm)                                             0.169354  ...
Sample Temperature                                    -0.006483  ...
% Moisture by Dry Weight of Cellulose Insulation      -0.013174  ...
% Relative Water Saturation in Oil                    -0.027618  ...
Moisture Content (ppm)                                -0.028312  ...
Neutralization Number (mg KOH/g)                       0.000818  ...
Inhibitor Content ()                                   0.042948  ...
Color                                                  0.015517  ...
Viscosity (cSt) at 25 °C                               0.021757  ...
Viscosity (SSU) at 25 °C                               0.020199  ...
Interfacial Tension (dynes/cm)                        -0.072912  ...
Power Factor at 25 °C                                 -0.013143  ...
Power Factor at 100 °C                                 0.003325  ...
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...    -0.010679  ...

                                                    Moisture Content
(ppm)  \
Measurement Number
0.088996
Tp                                                                -
0.035007
Ts
0.218712
Year                                                              -
0.331950
Impedance (kV)                                                    -
0.077189
Hydrogen
0.091707
Methane                                                           -
0.038066
Ethane                                                            -
0.027405
Ethylene                                                          -
0.026581
Acetylene                                                         -
0.028312
Carbon Monoxide
0.110689
Carbon Dioxide
0.183833
Nitrogen
0.071045
Oxygen                                                            -
0.118671
Total Dissolved Combustible Gases                                 -
0.030362
```

```
TDCG (ppm)
0.032751
Sample Temperature
0.219924
% Moisture by Dry Weight of Cellulose Insulation
0.724049
% Relative Water Saturation in Oil
0.833670
Moisture Content (ppm)
1.000000
Neutralization Number (mg KOH/g)
0.124470
Inhibitor Content ()
0.104928
Color
0.074133
Viscosity (cSt) at 25 °C                                        -
0.094932
Viscosity (SSU) at 25 °C                                        -
0.098365
Interfacial Tension (dynes/cm)                                  -
0.152444
Power Factor at 25 °C
0.068922
Power Factor at 100 °C
0.078236
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...              -
0.066738

                                                  Neutralization
Number (mg KOH/g)  \
Measurement Number
0.131683
Tp
-0.063318
Ts
0.096555
Year
-0.375845
Impedance (kV)
-0.054531
Hydrogen
0.020991
Methane
-0.015235
Ethane
-0.018092
Ethylene
0.035468
```

```
Acetylene
0.000818
Carbon Monoxide
0.030656
Carbon Dioxide
0.101048
Nitrogen
0.102577
Oxygen
0.034256
Total Dissolved Combustible Gases
0.070238
TDCG (ppm)
-0.005851
Sample Temperature
0.079240
% Moisture by Dry Weight of Cellulose Insulation
-0.021577
% Relative Water Saturation in Oil
0.046535
Moisture Content (ppm)
0.124470
Neutralization Number (mg KOH/g)
1.000000
Inhibitor Content ()
-0.056808
Color
0.260257
Viscosity (cSt) at 25 °C
-0.075087
Viscosity (SSU) at 25 °C
-0.075622
Interfacial Tension (dynes/cm)
-0.415786
Power Factor at 25 °C
0.227650
Power Factor at 100 °C
0.401280
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...
-0.022529

                                                    Inhibitor Content
()  \
Measurement Number                                                  -
0.028936
Tp                                                                  -
0.042587
Ts                                                                  -
0.070982
```

```
Year                                                    -
0.200802
Impedance (kV)                                          -
0.210278
Hydrogen
0.068996
Methane                                                 -
0.020633
Ethane                                                  -
0.162704
Ethylene
0.041194
Acetylene
0.042948
Carbon Monoxide                                         -
0.023879
Carbon Dioxide
0.033707
Nitrogen                                                -
0.077671
Oxygen
0.144401
Total Dissolved Combustible Gases                       -
0.030609
TDCG (ppm)                                              -
0.021596
Sample Temperature                                      -
0.018699
% Moisture by Dry Weight of Cellulose Insulation
0.187797
% Relative Water Saturation in Oil
0.142055
Moisture Content (ppm)
0.104928
Neutralization Number (mg KOH/g)                        -
0.056808
Inhibitor Content ()
1.000000
Color                                                   -
0.007393
Viscosity (cSt) at 25 °C                                -
0.252440
Viscosity (SSU) at 25 °C                                -
0.257743
Interfacial Tension (dynes/cm)
0.081185
Power Factor at 25 °C                                   -
0.064952
Power Factor at 100 °C                                  -
0.072390
```

```
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...
0.033257

                                                    Color  \
Measurement Number                                  0.102474
Tp                                                  0.015721
Ts                                                  0.014208
Year                                               -0.337180
Impedance (kV)                                      0.024532
Hydrogen                                           -0.031327
Methane                                             0.029960
Ethane                                              0.063606
Ethylene                                            0.069724
Acetylene                                           0.015517
Carbon Monoxide                                     0.063859
Carbon Dioxide                                      0.137364
Nitrogen                                            0.077236
Oxygen                                             -0.113298
Total Dissolved Combustible Gases                   0.031909
TDCG (ppm)                                         -0.004256
Sample Temperature                                  0.029694
% Moisture by Dry Weight of Cellulose Insulation    0.049685
% Relative Water Saturation in Oil                  0.049281
Moisture Content (ppm)                              0.074133
Neutralization Number (mg KOH/g)                    0.260257
Inhibitor Content ()                               -0.007393
Color                                               1.000000
Viscosity (cSt) at 25 °C                           -0.032785
Viscosity (SSU) at 25 °C                           -0.034813
Interfacial Tension (dynes/cm)                     -0.290459
Power Factor at 25 °C                               0.233247
Power Factor at 100 °C                              0.439821
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)... -0.011450

                                                    Viscosity (cSt) at
25 °C  \
Measurement Number
0.000214
Tp
0.185938
Ts                                                                  -
0.078983
Year
0.168518
Impedance (kV)
0.115061
Hydrogen
0.014048
Methane
```

```
0.029116
Ethane
0.044893
Ethylene                                                                -
0.014452
Acetylene
0.021757
Carbon Monoxide                                                         -
0.113196
Carbon Dioxide                                                          -
0.202674
Nitrogen                                                                -
0.022173
Oxygen                                                                  -
0.073979
Total Dissolved Combustible Gases                                       -
0.042034
TDCG (ppm)                                                              -
0.069125
Sample Temperature                                                      -
0.114286
% Moisture by Dry Weight of Cellulose Insulation                        -
0.046438
% Relative Water Saturation in Oil                                      -
0.063258
Moisture Content (ppm)                                                  -
0.094932
Neutralization Number (mg KOH/g)                                        -
0.075087
Inhibitor Content ()                                                    -
0.252440
Color                                                                   -
0.032785
Viscosity (cSt) at 25 °C
1.000000
Viscosity (SSU) at 25 °C
0.999727
Interfacial Tension (dynes/cm)
0.069606
Power Factor at 25 °C                                                   -
0.068282
Power Factor at 100 °C                                                  -
0.071091
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...
0.049719

                                                        Viscosity (SSU) at
25 °C  \
Measurement Number                                                      -
```

0.000611
Tp
0.184051
Ts -
0.079440
Year
0.173908
Impedance (kV)
0.112102
Hydrogen
0.012675
Methane
0.026225
Ethane
0.040317
Ethylene -
0.016951
Acetylene
0.020199
Carbon Monoxide -
0.111579
Carbon Dioxide -
0.202502
Nitrogen -
0.023855
Oxygen -
0.072678
Total Dissolved Combustible Gases -
0.044012
TDCG (ppm) -
0.076396
Sample Temperature -
0.115536
% Moisture by Dry Weight of Cellulose Insulation -
0.046771
% Relative Water Saturation in Oil -
0.064821
Moisture Content (ppm) -
0.098365
Neutralization Number (mg KOH/g) -
0.075622
Inhibitor Content () -
0.257743
Color -
0.034813
Viscosity (cSt) at 25 °C
0.999727
Viscosity (SSU) at 25 °C
1.000000

```
Interfacial Tension (dynes/cm)
0.074618
Power Factor at 25 °C                                                -
0.071535
Power Factor at 100 °C                                               -
0.074307
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...
0.050023

                                                          Interfacial
Tension (dynes/cm)  \
Measurement Number
-0.150736
Tp
0.219579
Ts
-0.004872
Year
0.359911
Impedance (kV)
0.152328
Hydrogen
-0.040029
Methane
-0.063843
Ethane
-0.159603
Ethylene
-0.104643
Acetylene
-0.072912
Carbon Monoxide
-0.038107
Carbon Dioxide
-0.184882
Nitrogen
0.009100
Oxygen
0.237519
Total Dissolved Combustible Gases
0.008186
TDCG (ppm)
0.008771
Sample Temperature
0.007414
% Moisture by Dry Weight of Cellulose Insulation
-0.173345
% Relative Water Saturation in Oil
-0.120141
```

```
Moisture Content (ppm)
-0.152444
Neutralization Number (mg KOH/g)
-0.415786
Inhibitor Content ()
0.081185
Color
-0.290459
Viscosity (cSt) at 25 °C
0.069606
Viscosity (SSU) at 25 °C
0.074618
Interfacial Tension (dynes/cm)
1.000000
Power Factor at 25 °C
-0.142200
Power Factor at 100 °C
-0.323377
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...
-0.026935

                                                    Power Factor at 25
°C  \
Measurement Number
0.030183
Tp
0.012373
Ts
0.013883
Year                                                                -
0.175599
Impedance (kV)
0.033291
Hydrogen
0.087803
Methane
0.047159
Ethane
0.054672
Ethylene
0.038914
Acetylene                                                           -
0.013143
Carbon Monoxide
0.091237
Carbon Dioxide
0.007155
Nitrogen
0.067922
```

```
Oxygen                                                                -
0.148743
Total Dissolved Combustible Gases
0.018729
TDCG (ppm)
0.014142
Sample Temperature
0.177078
% Moisture by Dry Weight of Cellulose Insulation                      -
0.072232
% Relative Water Saturation in Oil                                    -
0.013772
Moisture Content (ppm)
0.068922
Neutralization Number (mg KOH/g)
0.227650
Inhibitor Content ()                                                  -
0.064952
Color
0.233247
Viscosity (cSt) at 25 °C                                              -
0.068282
Viscosity (SSU) at 25 °C                                              -
0.071535
Interfacial Tension (dynes/cm)                                        -
0.142200
Power Factor at 25 °C
1.000000
Power Factor at 100 °C
0.309053
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...
0.018608

                                                        Power Factor at
100 °C  \
Measurement Number
0.096063
Tp
0.194837
Ts
0.102022
Year                                                                   -
0.412303
Impedance (kV)
0.280679
Hydrogen                                                               -
0.016694
Methane
0.051519
```

```
Ethane
0.032568
Ethylene
0.114889
Acetylene
0.003325
Carbon Monoxide
0.150220
Carbon Dioxide
0.119716
Nitrogen
0.015849
Oxygen                                                                -
0.105252
Total Dissolved Combustible Gases
0.015213
TDCG (ppm)
0.022761
Sample Temperature
0.021641
% Moisture by Dry Weight of Cellulose Insulation                      -
0.002901
% Relative Water Saturation in Oil                                    -
0.007252
Moisture Content (ppm)
0.078236
Neutralization Number (mg KOH/g)
0.401280
Inhibitor Content ()                                                  -
0.072390
Color
0.439821
Viscosity (cSt) at 25 °C                                              -
0.071091
Viscosity (SSU) at 25 °C                                              -
0.074307
Interfacial Tension (dynes/cm)                                        -
0.323377
Power Factor at 25 °C
0.309053
Power Factor at 100 °C
1.000000
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...                    -
0.012440

                                                        Dielectric
Breakdown (kV) at 1.016 mm (0.04 in) at 25 °C
Measurement Number
-0.013980
```

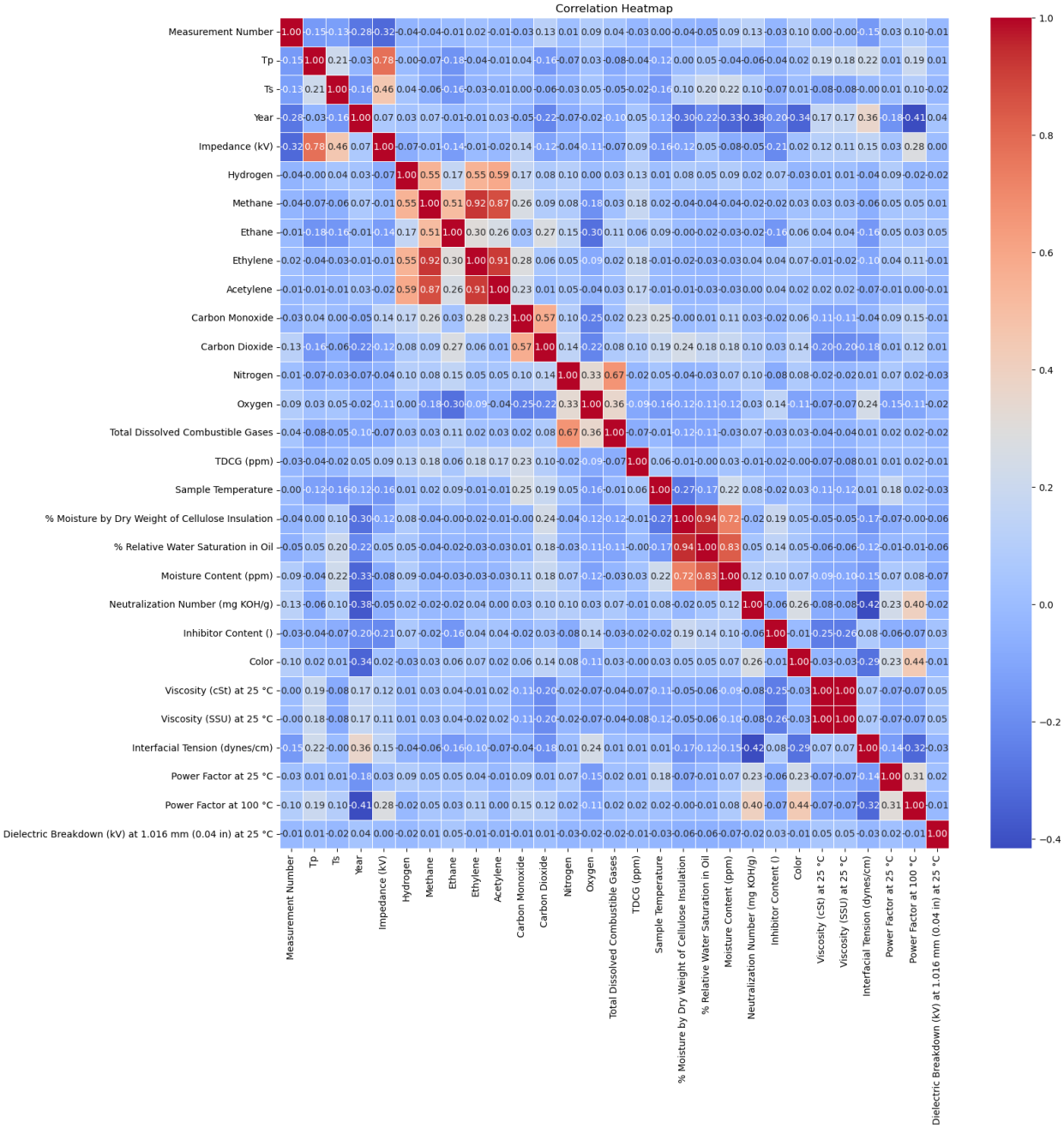
Tp
0.007745
Ts
-0.018111
Year
0.042264
Impedance (kV)
0.000164
Hydrogen
-0.015877
Methane
0.010466
Ethane
0.050282
Ethylene
-0.011526
Acetylene
-0.010679
Carbon Monoxide
-0.011977
Carbon Dioxide
0.012917
Nitrogen
-0.033365
Oxygen
-0.022750
Total Dissolved Combustible Gases
-0.021681
TDCG (ppm)
-0.009057
Sample Temperature
-0.033972
% Moisture by Dry Weight of Cellulose Insulation
-0.058308
% Relative Water Saturation in Oil
-0.057549
Moisture Content (ppm)
-0.066738
Neutralization Number (mg KOH/g)
-0.022529
Inhibitor Content ()
0.033257
Color
-0.011450
Viscosity (cSt) at 25 °C
0.049719
Viscosity (SSU) at 25 °C
0.050023
Interfacial Tension (dynes/cm)

```
-0.026935
Power Factor at 25 °C
0.018608
Power Factor at 100 °C
-0.012440
Dielectric Breakdown (kV) at 1.016 mm (0.04 in)...
1.000000

[29 rows x 29 columns]
```

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(16,16))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f",
linewidths=0.5)
plt.title("Correlation Heatmap")
plt.show()
```




```
def get_redundant_pairs(df):
    pairs_to_drop = set()
    cols = df.columns
    for i in range(0, df.shape[1]):
        for j in range(0, i+1):
            pairs_to_drop.add((cols[i], cols[j]))
    return pairs_to_drop

def get_top_abs_correlations(df, n=5):
    au_corr = df.corr().abs().unstack()
```

```
    labels_to_drop = get_redundant_pairs(df)
    au_corr =
au_corr.drop(labels=labels_to_drop).sort_values(ascending=False)
    return au_corr[0:n]

print("Top Correlations")
print(get_top_abs_correlations(corr, 50))
```

```
Top Correlations
Viscosity (cSt) at 25 °C                                Viscosity (SSU) at
25 °C                                0.999959
Ethylene                                                Acetylene
0.989840
% Moisture by Dry Weight of Cellulose Insulation  % Relative Water
Saturation in Oil                    0.984844
Methane                                                 Ethylene
0.981891
                                                        Acetylene
0.972914
% Relative Water Saturation in Oil                      Moisture Content
(ppm)                                0.929896
Tp                                                      Impedance (kV)
0.920439
Nitrogen                                                Total Dissolved
Combustible Gases                     0.913466
% Moisture by Dry Weight of Cellulose Insulation  Moisture Content
(ppm)                                0.902997
Hydrogen                                                Acetylene
0.865899
                                                        Ethylene
0.842037
                                                        Methane
0.829283
Color                                                   Power Factor at 100
°C                                   0.758568
Neutralization Number (mg KOH/g)                        Interfacial Tension
(dynes/cm)                           0.745358
Carbon Monoxide                                         Carbon Dioxide
0.729788
Neutralization Number (mg KOH/g)                        Power Factor at 100
°C                                   0.705981
Year                                                    Neutralization
Number (mg KOH/g)                      0.702773
Methane                                                 Ethane
0.695704
Year                                                    Interfacial Tension
(dynes/cm)                           0.686419
                                                        Power Factor at 100
°C                                   0.670720
Color                                                   Interfacial Tension
```

(dynes/cm) 0.661088
Year Color
0.654922
Interfacial Tension (dynes/cm) Power Factor at 100
°C 0.637173
Ts Impedance (kV)
0.617701
Neutralization Number (mg KOH/g) Color
0.616162
Year Moisture Content
(ppm) 0.584501
Power Factor at 25 °C Power Factor at 100
°C 0.583098
Ethane Ethylene
0.579657
Oxygen Total Dissolved
Combustible Gases 0.565751
Ethane Acetylene
0.539331
Measurement Number Impedance (kV)
0.527770
Color Power Factor at 25
°C 0.512165
Neutralization Number (mg KOH/g) Power Factor at 25
°C 0.504669
Carbon Monoxide Oxygen
0.492170
Inhibitor Content () Viscosity (SSU) at
25 °C 0.491973
Viscosity (cSt) at
25 °C 0.490764
Carbon Dioxide Interfacial Tension
(dynes/cm) 0.488797
Nitrogen Oxygen
0.484964
Tp Ts
0.481511
Carbon Dioxide Viscosity (SSU) at
25 °C 0.477290
Viscosity (cSt) at
25 °C 0.476894
Ethane Oxygen
0.474895
Year % Moisture by Dry
Weight of Cellulose Insulation 0.461460
Measurement Number Year
0.458742
Year Carbon Dioxide
0.451708

```
Hydrogen                                                Ethane
0.444478
Carbon Monoxide                                         Sample Temperature
0.444355
Tp                                                      Ethane
0.440523
Year                                                    % Relative Water
Saturation in Oil                  0.437868
Interfacial Tension (dynes/cm)                          Power Factor at 25
°C                                 0.436669
dtype: float64
```

Notes: Any correlation above 0.5 is considered highly correlated and may be reasonably imputated through PCU.